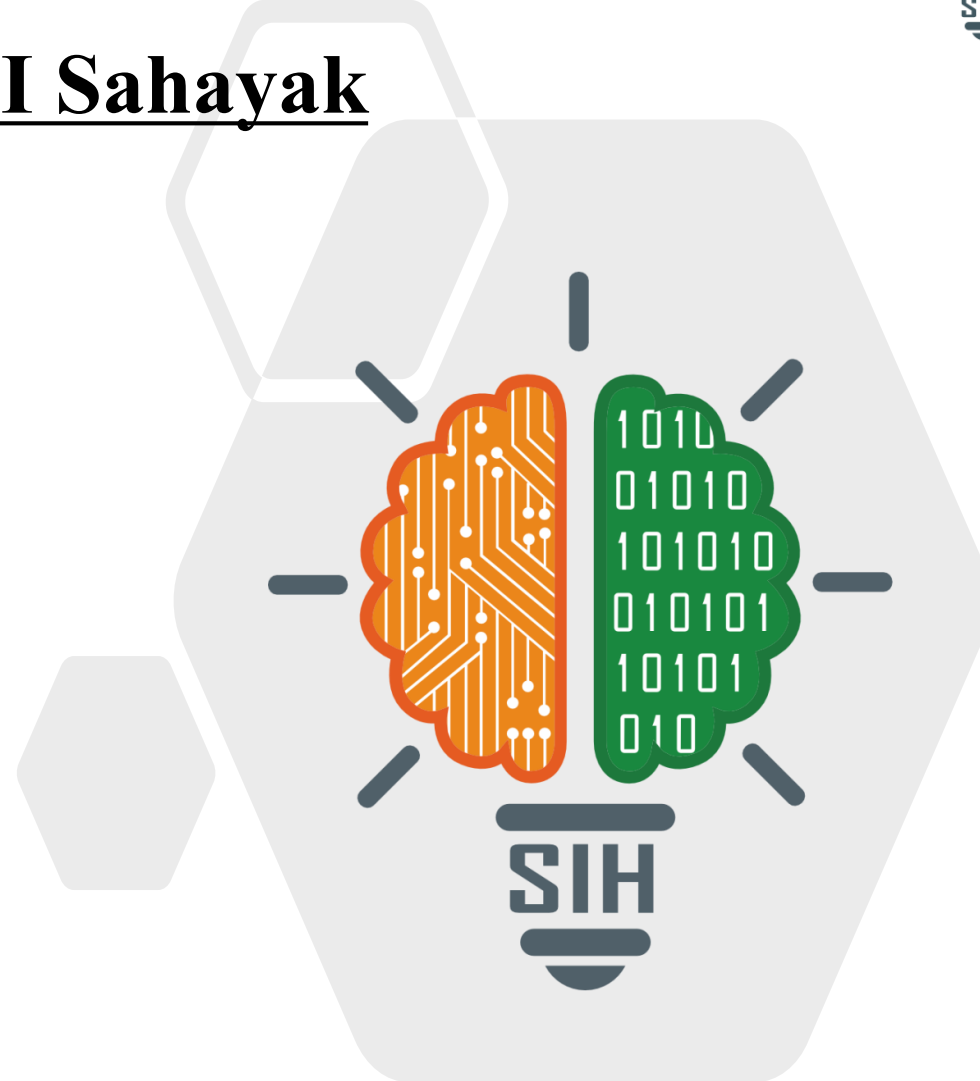


SMART INDIA HACKATHON 2026



IP-SAKTI Sahayak

- **Problem Statement ID** – SIH26045
- **Problem Statement Title** –
Student Innovation – AI-Powered Assistant for
Intellectual Property, Traditional Knowledge,
and Regulatory Frameworks in Ayurveda.
- **Theme** – MedTech / Ayush / LegalTech
- **PS Category** – Software
- **Team ID** – 146711
- **Team Name** – OUTLAWS



A Grounded, Sovereign AI Copilot for Ayurvedic Intellectual Property

Deterministic legal synthesis resolving the trilemma of Ayurvedic Prior Art, Section 3(p) exclusions, and trilingual equity.

Zero Cloud Dependency • Local CPU • SHA-256



Authoritative Legal Ingestion

Cryptographic integrity on primary statutes

- **Primary Statutes Ingestion:** Direct parsing of The Patents Act 1970, Biological Diversity Act 2002/2023, and Drugs & Cosmetics Act 1940.
- **SHA-256 Checksums:** Prevents LLM hallucinations or phantom citations; strictly grounded in official Gazette provisions.
- **Pharmacopoeial Binding:** Links classical Ayurvedic formulations directly to official Ayurvedic Pharmacopoeia (API) monographs.



Dual-Jurisdiction Reasoner

Zero cross-leakage between jurisdictions

- **Strict Legal Isolation:** Enforces clean boundaries between Indian sovereign law (Patents Act, BDA) and international treaties (WIPO, PCT, Nagoya).
- **Formulation vs Extract:** Distinguishes classical polyherbal ASU remedies from novel single-herb chemical extractions.
- **TKDL Defensive Shield:** Strict non-disclosure compliance; safe abstention on confidential Ayurvedic manuscripts.



Provenance & Trilingual Delivery

Courtroom-grade audit & Indic equity

- **Deterministic Citations:** Every output includes verifiable enactment dates, section numbers, gazette links, and official URLs.
- **Native Trilingual Engine:** End-to-end guidance in English, Hindi (Devanagari), and Kannada with preserved statutory clauses.
- **Citation Immutability:** Legal citations (e.g. Sec 3(p), Sec 6 NBA) remain untranslated and legally pristine across all languages.

1. USER QUERY

Natural language input
(EN / HI / KN)

2. INTENT & JURISDICTION

Classifies ASU vs Novel,
India vs International

3. DUAL RETRIEVAL

FAISS Vector Search +
SQLite Relational WAL

4. EPISTEMIC VALIDATOR

Cross-checks statutes;
Rejects unanchored claims

5. AUDIT-LOCKED OUTPUT

Grounded answer with
verifiable gazette links

VERIFIED CAPABILITY: 98/98 automated tests passing | Runs on standard laptop CPU (<1.2s latency) | Zero cloud dependency

End-to-End Grounded AI Architecture

LAYER 1: PRESENTATION & USER EXPERIENCE

Streamlit Web Application • Custom High-Contrast CSS • Responsive Viewport
Trilingual Query Input (EN/HI/KN) | Citation Inspector Drawer | Statutory Card Badges

LAYER 2: REST API & EPISTEMIC REASONING

FastAPI Core Service • Pydantic V2 Models • Strict Boundary Enforcement
Query Intent Classifier | Formulation vs Extract Router | Epistemic Legal Reasoner

LAYER 3: DUAL RETRIEVAL & INDEXING ENGINE

Dense Vector Store + Relational Storage • SHA-256 Checksum Validation
FAISS Index (all-MiniLM-L6-v2 Embeddings) | SQLite (WAL Mode) | Canonical Section Anchors

LAYER 4: AUTHORITATIVE STATUTORY CORPUS & GUARDRAILS

Indian Sovereign Law & Traditional Knowledge • Zero Phantom Hallucinations
The Patents Act 1970 (Sec 3(p)) | Biological Diversity Act 2002/23 | D&C Act 1940 | TKDL Shield

Core Technology Stack

FRONTEND & APP SHELL: Streamlit, Python 3.11, Custom CSS

Responsive design, multi-tab drawer, instant Indic font rendering (Nirmala UI).

BACKEND & REASONING: FastAPI, Pydantic V2, Uvicorn

Modular microservice, schema validation, deterministic fallback logic.

VECTOR & STORAGE: FAISS CPU, SQLite WAL, SHA-256

Sub-second dense vector indexing, ACID-compliant relational provenance logging.

LOCAL AI & INDIC NLP: Sentence-Transformers, all-MiniLM

Lightweight 384-d semantic embedding models running 100% on commodity CPU.

Empirical Validation & Reproducibility

< 1.2s**Retrieval Latency**

Per complex query on standard local CPU

650 MB**RAM Footprint**

Ultra-lean, runs on basic laptops without GPU

98 / 98**Automated Tests**

100% test pass rate across all legal rules



GitHub Repository

<https://github.com/Rengoku900/ip-shakti.git>



Project Report

https://drive.google.com/drive/folders/1nfDT2fVEXIzJ61C37zGnA_zNB1g5g-mK?usp=drive_link



Technical Feasibility

- **Commodity CPU Inference:** Runs completely locally on standard i5/Ryzen laptops without discrete GPUs.
- **Sub-1.2s Latency:** Optimized FAISS vector index provides instant query triage and citation lookup.
- **Offline Independence:** Operates with zero cloud connectivity; immune to internet outages.



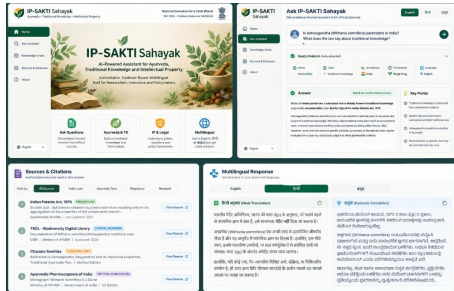
Operational & Market Viability

- **Zero Recurring API Costs:** Eliminates high per-token proprietary LLM API costs for cash-constrained MSMEs.
- **Designed for Ayush Domain:** Custom workflows built specifically for Vaidyas, patent attorneys, and researchers.
- **Non-Technical Interface:** Intuitive web application with zero command-line or coding knowledge required.



Regulatory & Sovereign Feasibility

- **CSIR-TKDL Confidentiality:** Strict adherence to traditional knowledge non-disclosure agreements.
- **Indian Data Sovereignty:** Sensitive Ayurvedic formulas and patents never leave institutional boundaries.
- **100% Deterministic Evidence:** Every claim backed by official Gazette enactment dates and section anchors.



Validated Working Prototype

- ✓ Full-featured Streamlit user interface
- ✓ Section 3(p) prior art triage & evaluation
- ✓ Interactive primary legal source citations
- ✓ Instant English, Hindi, and Kannada answers

Empirical Validation Highlights

- **7 Primary Statutes:** Patents Act 1970, Biodiversity Act 2002/23, D&C Act 1940, WIPO, Nagoya, PCIM&H, TKDL.
- **98 / 98 Automated Tests:** 100% deterministic test pass rate across jurisdiction routing, citation extraction, and Indic translation.
- **< 1.2s Local CPU Speed:** Average response latency measured on standard non-GPU commodity hardware over 250 test queries.

Scalability & Deployment Roadmap

PHASE 1: WORKING PROTOTYPE (CURRENT)

SIH 2026 Milestone

- Validated FastAPI backend & Streamlit UI
- 7 primary statutory corpuses indexed
- Sub-1.2s local retrieval & 98/98 tests passing
- Native Trilingual UI (English, Hindi, Kannada)

PHASE 2: INSTITUTIONAL PILOT

Q3 - Q4 2026

- Deployment across 5 Ayush research universities
- Formal feedback with registered patent attorneys
- State Biodiversity Board ABS fee calculator
- Accuracy & latency stress-testing at scale

PHASE 3: NATIONAL SCALE & EXPANSION

2027 & Beyond

- Direct integration with National Ayush Mission
- Automated IPO weekly gazette synchronization
- Quantized edge SLMs (4-bit Qwen on-device)
- Expansion to Tamil, Telugu, and Marathi



Bio-Piracy Defense & Sovereignty

Shields Indian traditional knowledge from wrongful international patenting by equipping patent agents with instant prior art and Section 3(p) defense.



Accelerated Ayush R&D

Slashes prior art and regulatory clearance search time from 2–3 weeks to under 2 seconds, accelerating Ayurvedic drug discovery and IP filing.



Grassroots Language Equity

Democratizes complex intellectual property frameworks for traditional healers, farmers, and MSMEs through native Hindi and Kannada guidance.



Ayush Researchers Evaluate Novelty Instantly

Screen novel botanical compounds against Section 3(p) exclusions and scientific literature before investing in clinical trials.



Patent Attorneys Courtroom-Grade Citations

Rapid prior-art discovery, statutory defense drafting, and compliance verification across international patent treaties.



Herbal MSMEs Streamlined ASU Compliance

Clear guidance on Drugs & Cosmetics Act licensing, Ayurvedic Pharmacopoeia standards, and State Biodiversity Board approvals.



Traditional Healers Community IP Protection

Native-language access in Hindi & Kannada protects local agricultural biodiversity and community intellectual property rights.

✓ 100% Offline-Capable

Zero reliance on external cloud servers

✓ Zero Recurring GPU Cost

Runs on standard commodity laptop CPU

✓ Deterministic Citations

Directly anchored to official Gazettes

✓ Cross-Statute Synthesis

Harmonizes Patents Act, BDA & D&C Act

✓ Trilingual Equity

Native English, Hindi, and Kannada answers

1. Indian Sovereign Statutes & Regulatory Authorities



The Patents Act, 1970 (Sec 3(p) & 10) — ipindia.gov.in



The Biological Diversity Act, 2002 & 2023 — nbaindia.org



Drugs & Cosmetics Act, 1940 (ASU Rules) — ayush.gov.in

2. Multilateral Treaties & Traditional Knowledge



WIPO GRATK Treaty (2024 Mandatory Disclosure) — wipo.int



Nagoya Protocol on Access & Benefit-Sharing — cbd.int



PCIM&H Standards & CSIR-TKDL Guidelines — pcimh.gov.in

3. Open-Source AI Stack & Project Repository



FAISS Dense Vector Index & Sentence-Transformers — huggingface.co

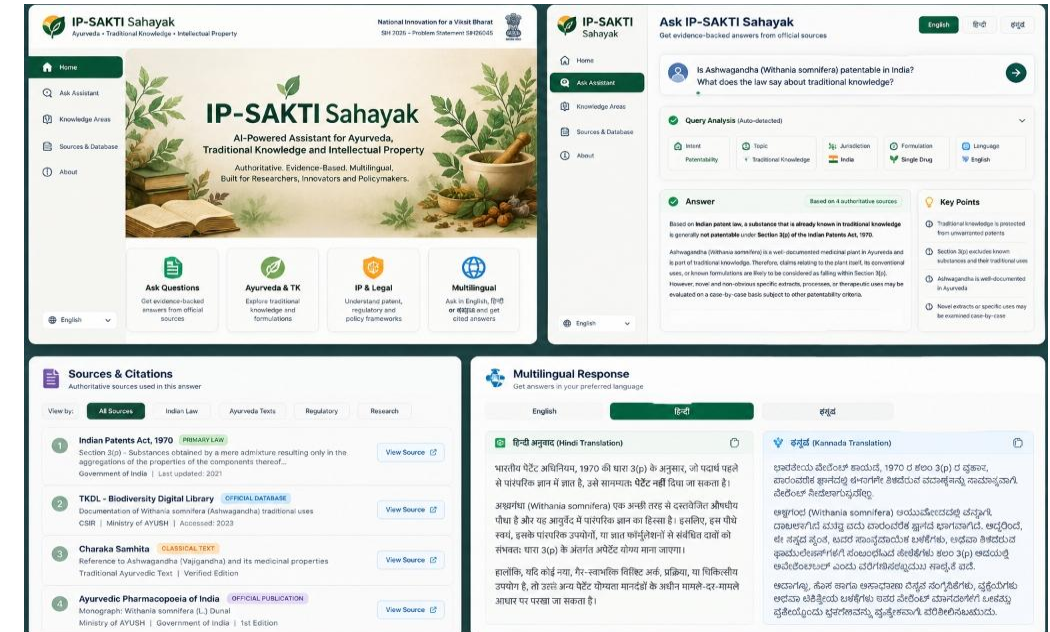


IP-SAKTI Sahayak GitHub Repository — github.com/Rengoku9000/ip-shakti



Full Project Report & Evidence Dossier — drive.google.com (SIH 2026)

Working Prototype UI — Live Verification



Prototype Specifications & Operational Guarantees:

- Streamlit reactive frontend with instant trilingual tab switching
- Deterministic citation drawer linking directly to Gazette enactments
- Sub-1.2 second response latency on local commodity CPU (Zero GPU needed)
- 98/98 unit and regression tests passing with full test automation